

SpaceX Falcon 9 First Stage Landing Prediction

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

**IBM Data Science Professional Certificate
Capstone Project**

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

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EXECUTIVE SUMMARY



- **Objective**
 - Predict Falcon 9 first-stage landing success using historical launch data.
- **Methodology Summary**
 - Data Collection through **SpaceX REST API**
 - Data Collection with **Web Scraping (Wikipedia)**
 - Data Wrangling
 - Exploratory Data Analysis (**Visualization & SQL**)
 - Interactive Visual Analytics (**Folium & Plotly Dash**)
 - Predictive Analysis using Machine Learning
- **Key Results**
 - Landing success increased over time.
 - Payload mass, orbit, and launch site significantly affected landing success.
 - The best machine learning model achieved the highest prediction accuracy.

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INTRODUCTION



- **Project Background:** SpaceX reduces launch costs by successfully recovering and reusing Falcon 9 first-stage boosters, making launches more efficient and cost-effective.
- **Problem Statement:** Predict whether the Falcon 9 first-stage landing will be successful before launch using historical mission data and machine learning techniques.
- **Project Objective:** Build, train, and evaluate machine learning classification models to predict Falcon 9 first-stage landing success using historical launch data.
- **Data Sources:**
 - **SpaceX REST API:** Launch information, payload details, booster version, launch site, and mission outcomes.
 - **Wikipedia (Web Scraping):** Additional Falcon 9 launch records used to validate and enrich the dataset.



METHODOLOGY



- **Data Collection:**
 - Collected Falcon 9 launch data using the **SpaceX REST API**.
 - Collected additional launch records through **Wikipedia Web Scraping**.
- **Data Wrangling:**
 - Cleaned missing values and standardized the dataset.
 - Merged API and Wikipedia datasets.
- **Exploratory Data Analysis (EDA):**
 - Created visualizations using Matplotlib and Seaborn.
 - Analyzed relationships between mission features and landing success.
- **SQL Analysis:**
 - Queried the database to analyze launch statistics.
 - Retrieved insights using SQL queries.
- **Interactive Visualization:**
 - Built interactive **Folium** maps.
 - Developed a **Plotly Dash** dashboard.
- **Predictive Analysis:**
 - Built and evaluated multiple classification models.
 - Selected the best-performing model.



EDA Using SQL Queries (Part 1)

SQL Query Output – Launch Site Analysis

```
%%sql
```

```
SELECT Launch_Site,  
COUNT(*) AS Total_Launches  
FROM SPACEXTABLE  
GROUP BY Launch_Site;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site	Total_Launches
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

- **Key Findings:**
- Used SQL to calculate total launches for each Falcon 9 launch site.
- **CCAFS SLC-40** recorded the highest number of launches (**34**).
- **CCAFS LC-40** completed **26** launches during the analyzed period.
- KSC LC-39A completed 25 launches.
- **VAFB SLC-4E** recorded the fewest launches (**16**).
- These SQL results identified the busiest launch sites and supported subsequent visualization and predictive analysis.



EDA Using SQL Queries (Part 2)

Mission Outcome Analysis (SQL Query Result)

```
%%sql

SELECT Mission_Outcome,
       COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Mission_Outcome
ORDER BY Count DESC;

* sqlite:///my_data1.db
Done.
```

Mission_Outcome	Count
Success	30
Failure (in flight)	1

- **Key Findings:**
- SQL was used to summarize mission outcomes.
- **30 missions** were completed successfully.
- Only **1 mission** resulted in failure during the selected period.
- The high success rate indicates improved Falcon 9 reliability.
- SQL aggregation helped validate mission performance before predictive modeling.

Results

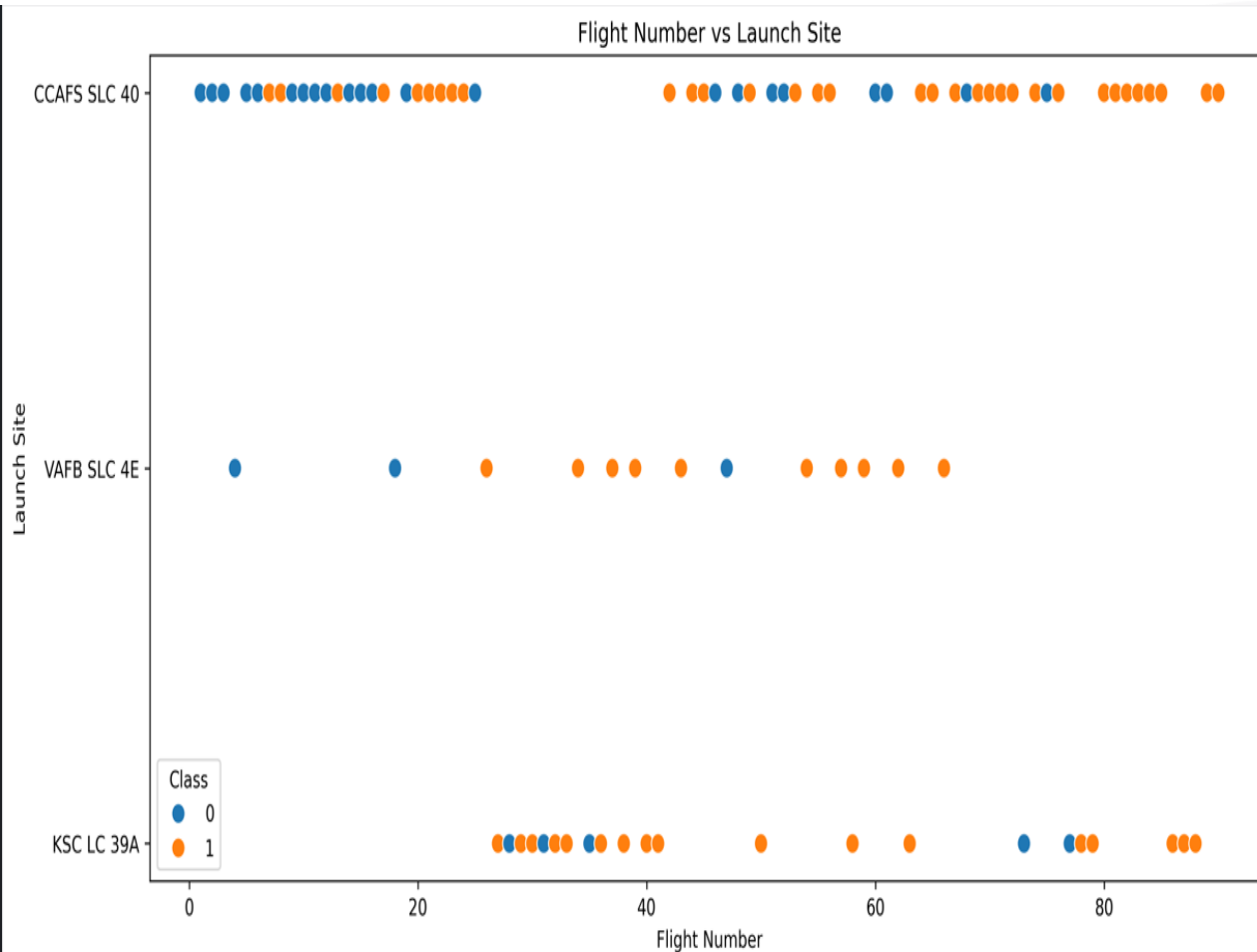
- **Landing Success Trend:** Landing success improved steadily over time.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis:** The best classification model achieved the highest prediction accuracy.
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 first-stage landing success and support mission planning.

GitHub Repository:

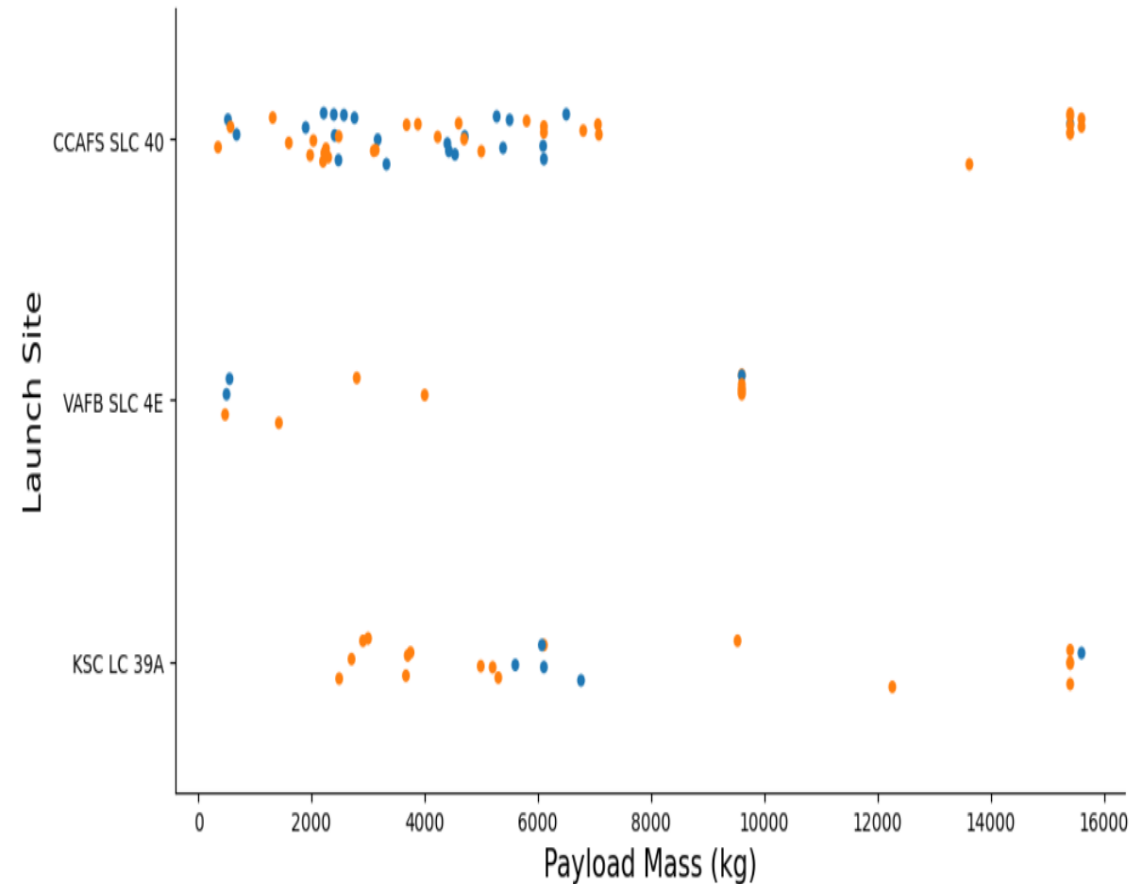
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Flight Number vs. Launch Site



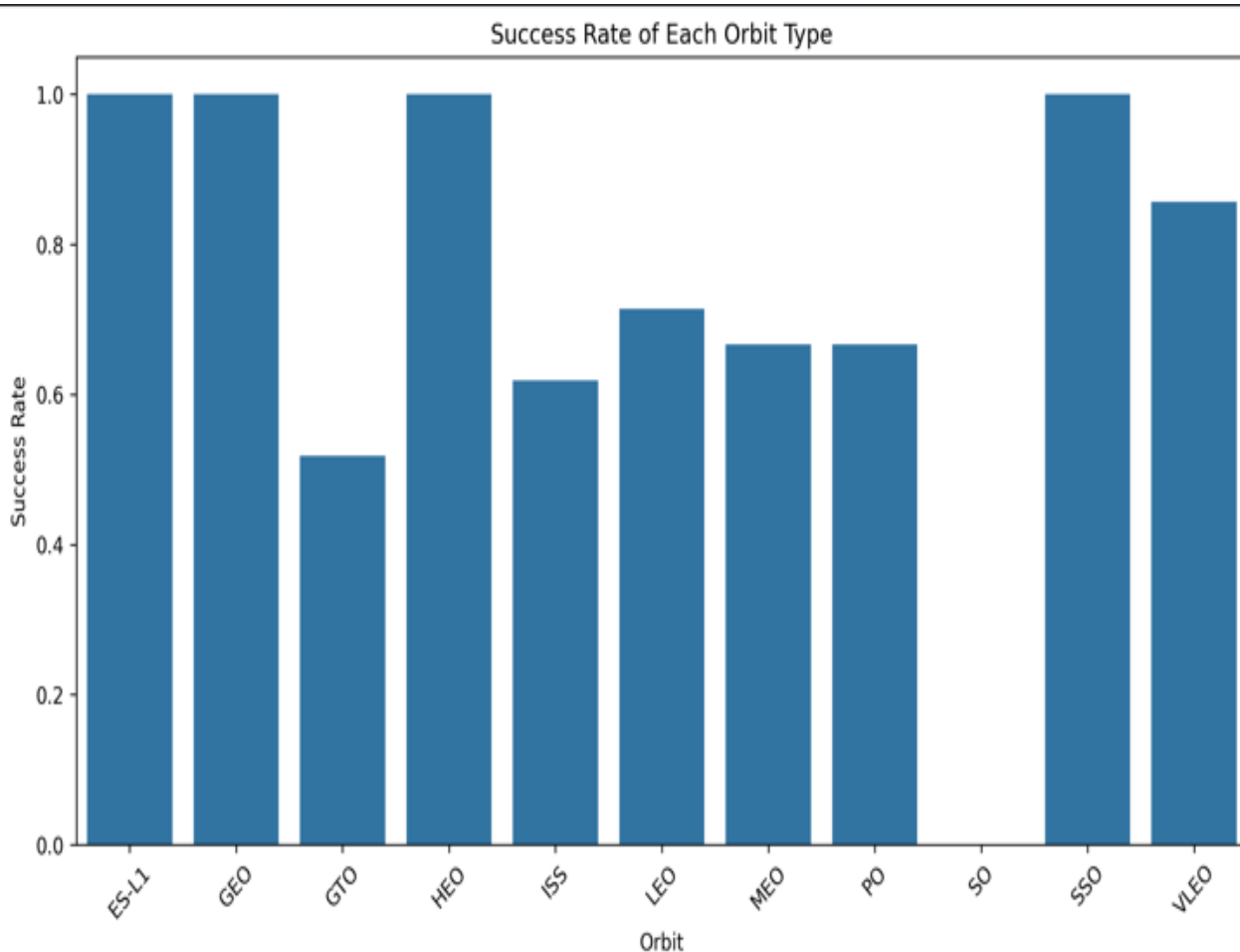
Payload Mass vs. Launch Site



- **Key Findings**

- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, launch site and payload mass together influenced Falcon 9 landing success.

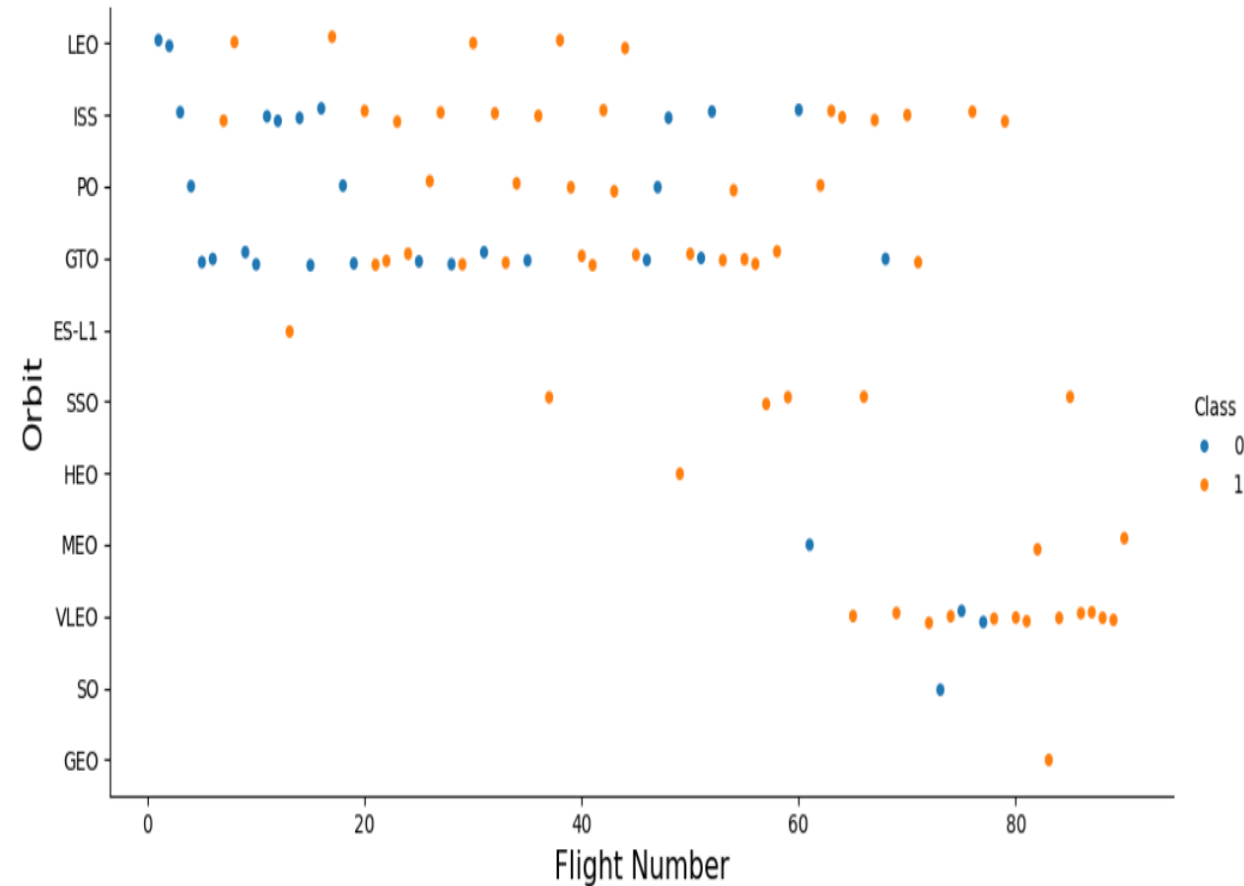
Success Rate by Orbit



• Key Findings

- **ES-L1, GEO, HEO, and SSO** achieved a **100% landing success rate**, indicating highly reliable mission outcomes for these orbit types.
- **VLEO** also demonstrated a high landing success rate of approximately **86%**, showing strong booster recovery performance.
- LEO, MEO, ISS, and PO achieved moderate landing success rates, indicating variable mission outcomes across these orbit types.
- **GTO** recorded a comparatively lower landing success rate, while **SO** showed **no successful landings** in the available dataset.
- Overall, orbit type significantly influenced Falcon 9 landing success, although mission characteristics and operational factors also played important roles.

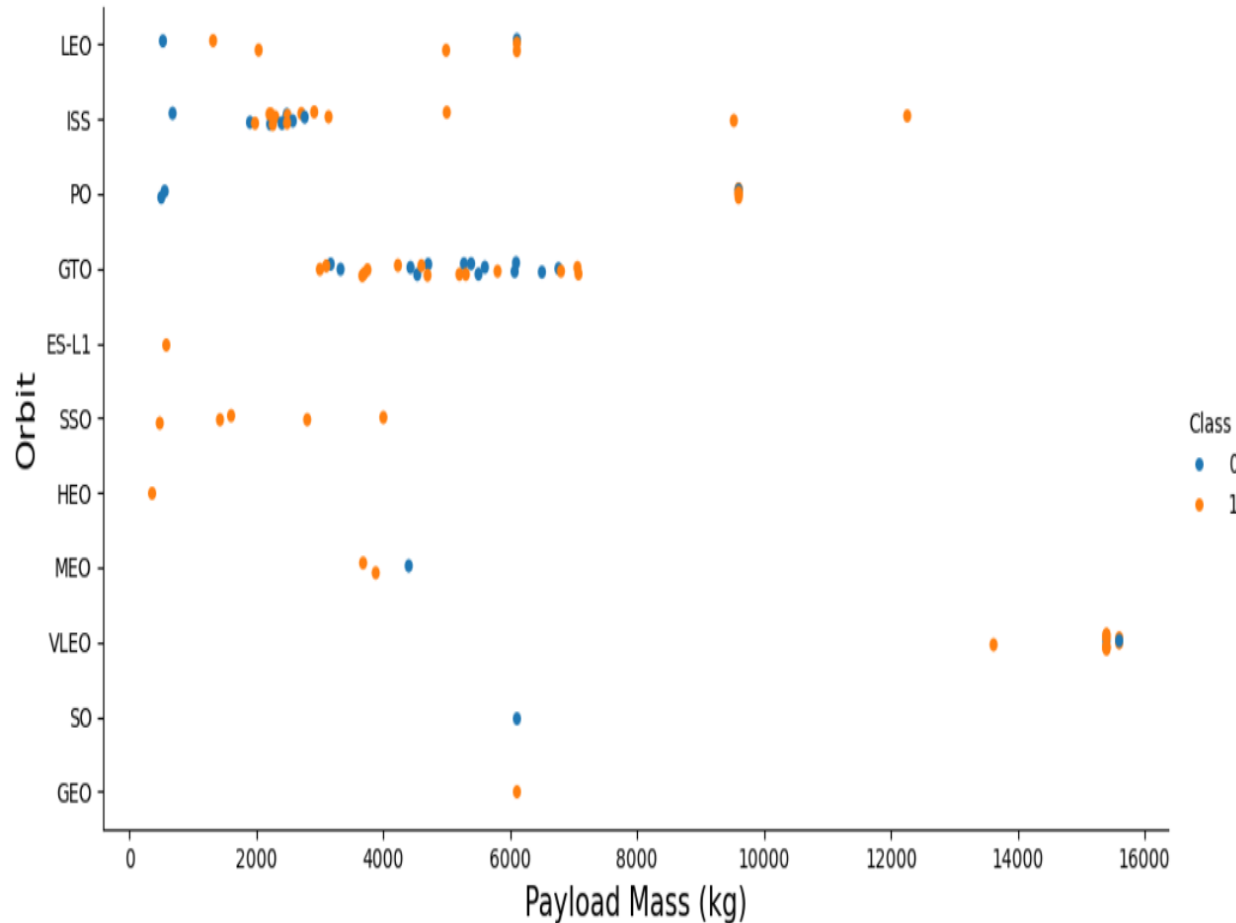
Flight Number vs. Orbit



• Key Findings

- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Orbit type influenced landing success, while payload mass and launch site also played important roles.

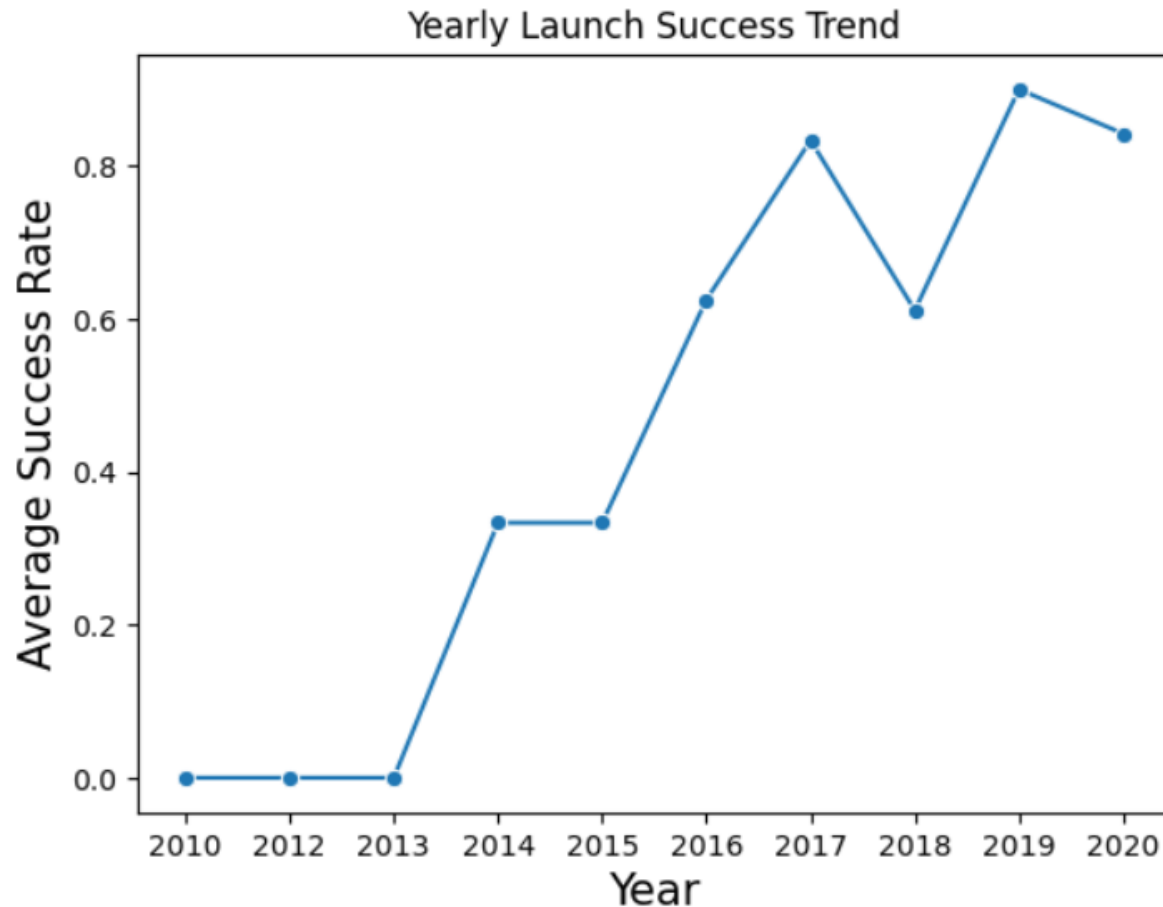
Payload Mass vs. Orbit



• Key Findings

- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low to medium payload masses.
- Payload mass alone did not determine Falcon 9 landing success; orbit type and mission characteristics also influenced landing outcomes.

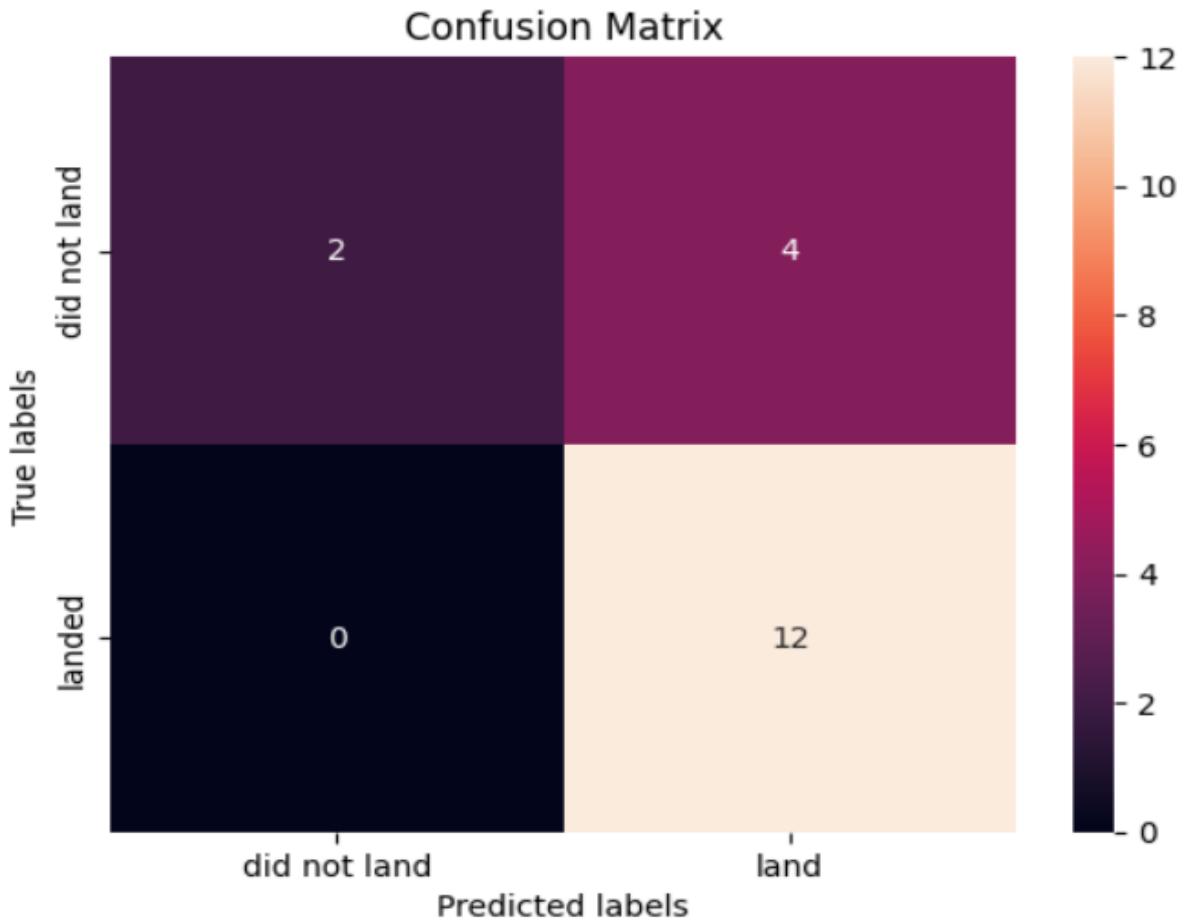
Yearly Launch Success Trend



• Key Findings

- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onward, average launch success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high launch success rates, demonstrating improved booster recovery and mission reliability.

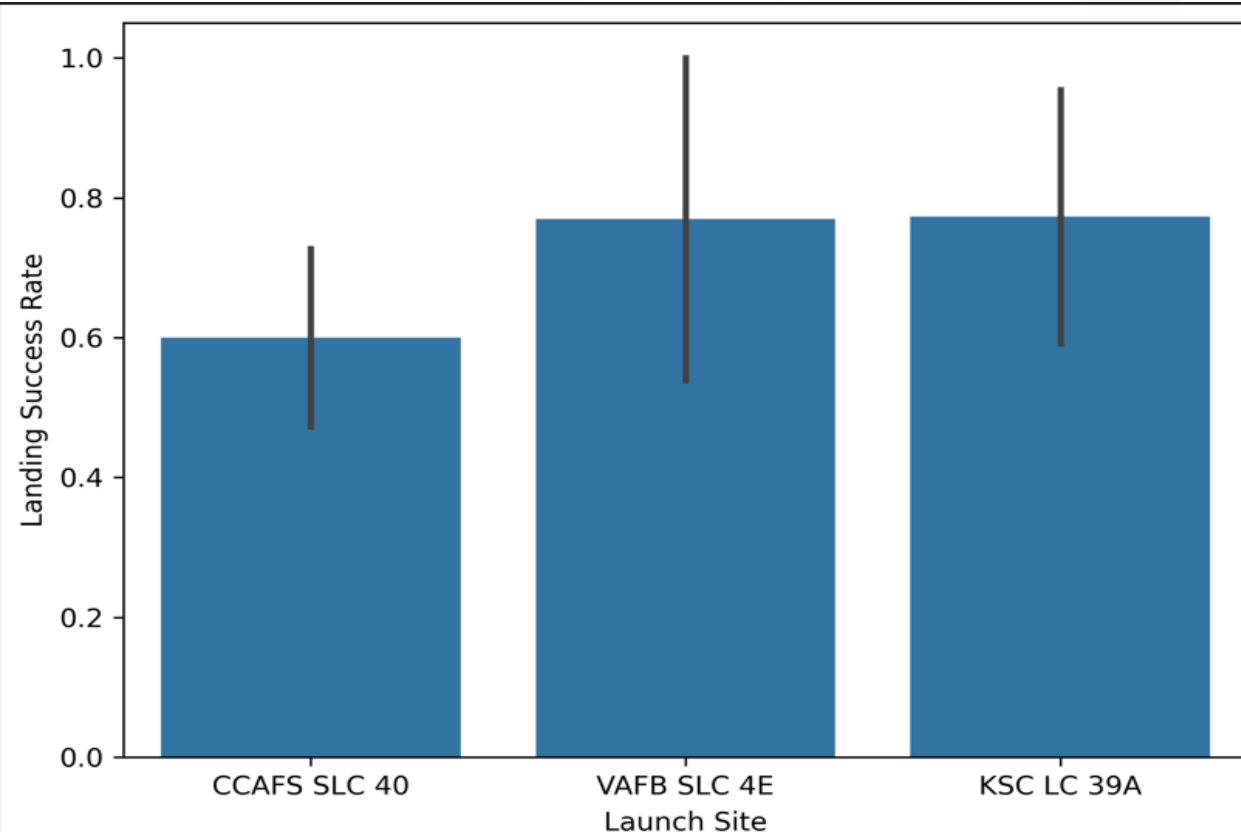
Model Evaluation – Confusion Matrix



- **Key Findings**

- The model correctly predicted **12 successful landings (True Positives)**.
- **2 unsuccessful landings** were correctly classified (True Negatives).
- **4 unsuccessful landings** were incorrectly predicted as successful (False Positives).
- The model demonstrated strong performance in identifying successful landings, with no false negatives, although some unsuccessful missions were incorrectly classified as successful.

Landing Success by Launch Site



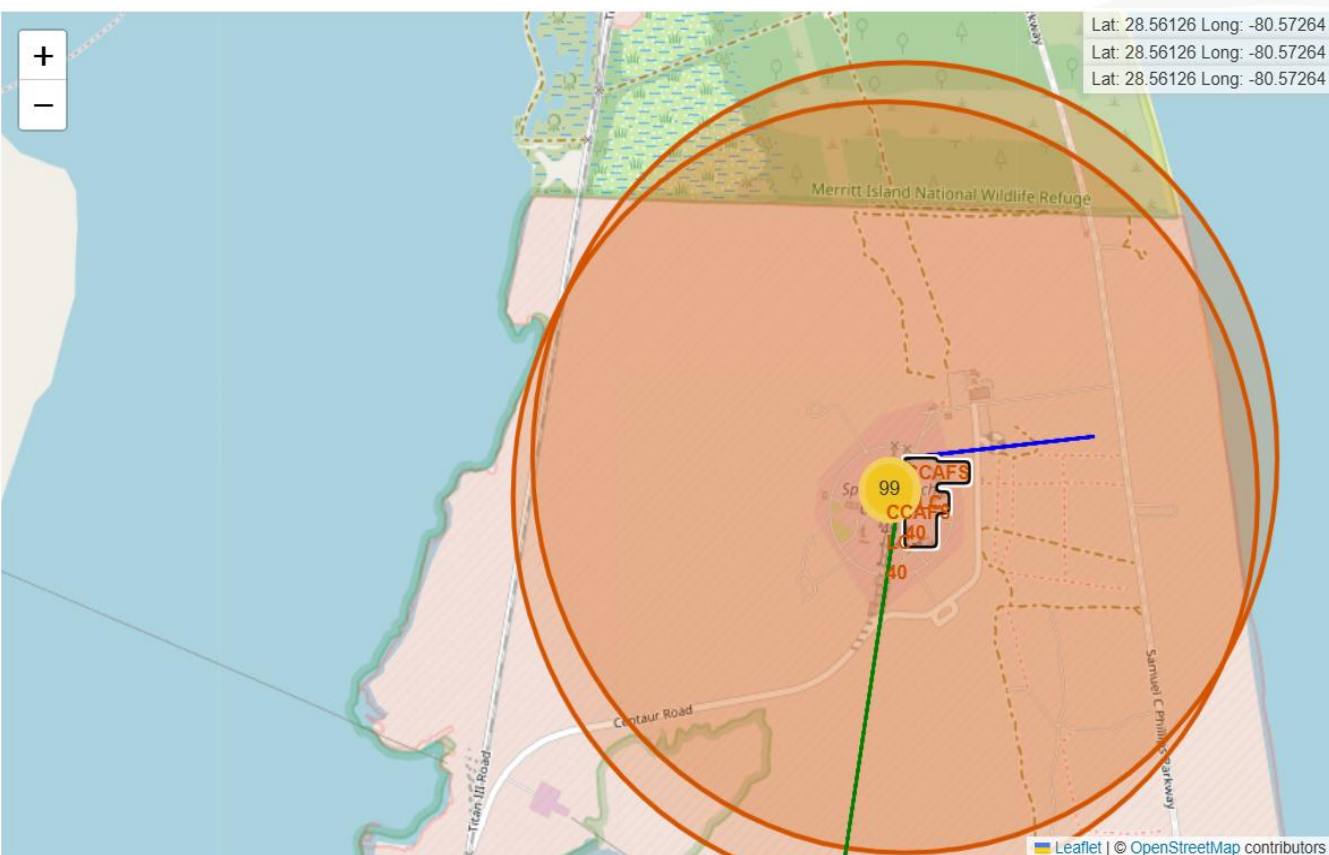
- **Key Findings**
- **KSC LC 39A** achieved the **highest average landing success rate** among all launch sites.
- **VAFB SLC 4E** also recorded a **high landing success rate**, indicating strong mission performance.
- **CCAFS SLC 40** showed the **lowest average landing success rate** in the dataset.
- Launch site significantly influenced Falcon 9 first-stage landing success.

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Launch Site Proximity Analysis (CCAFS)



• Key Findings:

- CCAFS launch site is located close to the coastline.
- The coastal location supports safe launch trajectories over the ocean.
- The site is positioned away from major populated areas to improve safety.
- The proximity map highlights CCAFS's strategic coastal location for safe launch operations.

SPACEX FALCON 9 – FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved over time.
- Launch site, payload mass, flight number, and orbit type significantly influenced landing success.
- The best-performing classification model demonstrated strong predictive performance.

Implications

- Machine learning can effectively predict Falcon 9 landing success.
- The predictive model can support mission planning and reduce launch costs.
- Data-driven insights help improve future SpaceX launch decisions.



Results – Visualizations & Dashboard

EDA Visualizations

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Launch Success Trend
- Landing Success by Launch Site

Interactive Dashboard

- SQL Query Results
- Folium Launch Site Map
- Plotly Dash Dashboard
- Confusion Matrix (Model Evaluation)

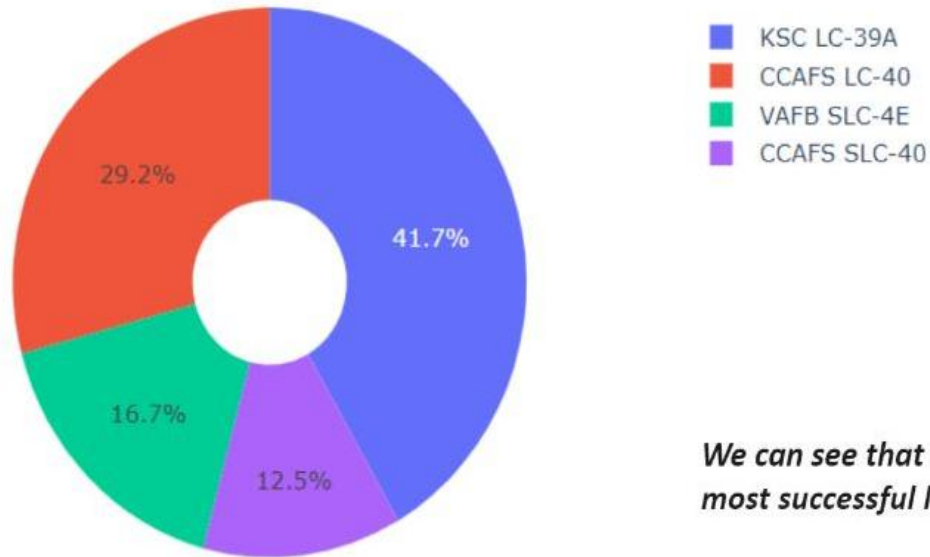
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Interactive Dashboard (Plotly Dash)

Total Success Launches by All Sites (Pie Chart)

Total Success Launches By all sites



We can see that KSC LC-39A had the most successful launches from all the sites

• Key Findings:

- Interactive dashboard developed using Plotly Dash.
- Displays successful launches across all launch sites.
- KSC LC-39A recorded the highest number of successful launches.

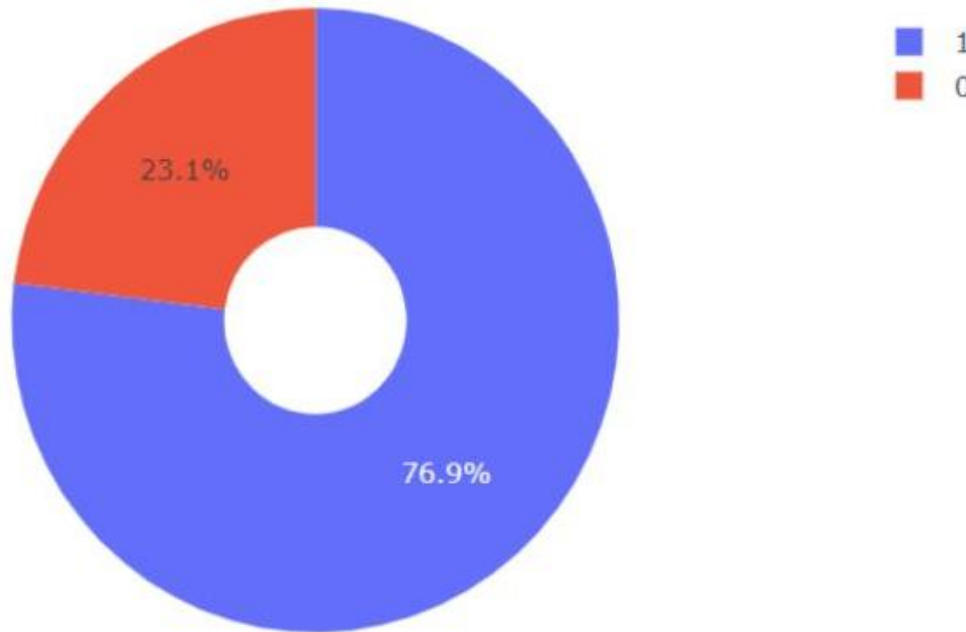
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Launch Site Analysis

KSC LC-39A: Success vs Failure



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

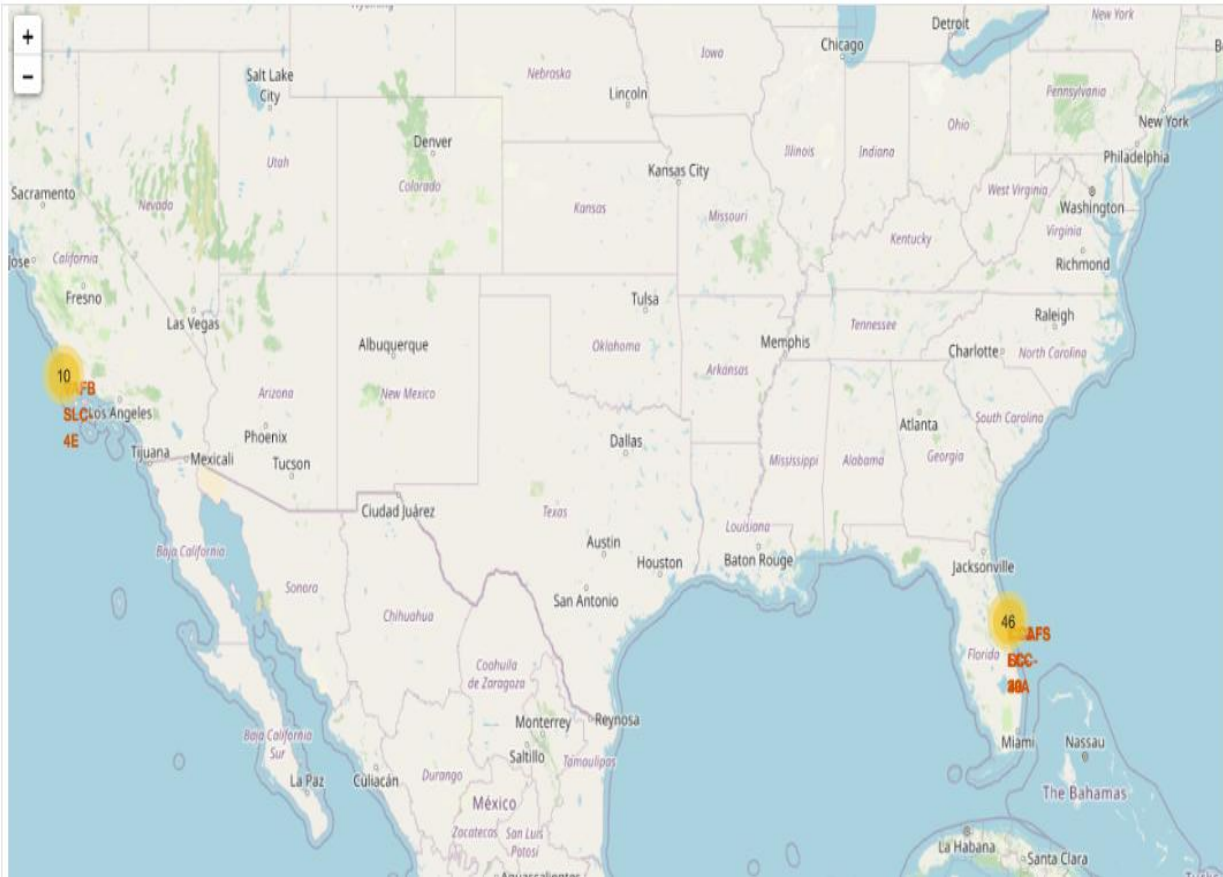
- **Key Findings:**
- Payload mass is compared with launch success.
- Different colors represent booster versions.
- The dashboard allows interactive payload range filtering.
- The visualization helps identify payload-related launch patterns.

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Interactive Launch Site Map (Folium)

Launch Site Distribution (Folium)

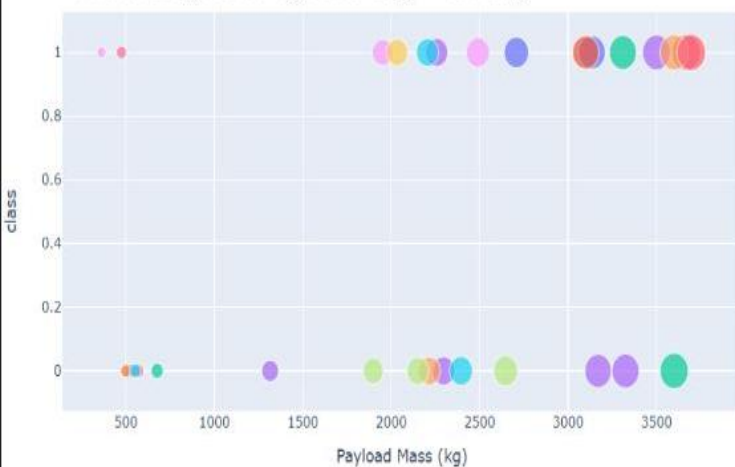


- **Key Findings:**
- Visualized Falcon 9 launch sites using Folium.
- Interactive markers display launch site locations and launch counts.
- CCAFS SLC-40 recorded the highest number of Falcon 9 launches, indicating it was the most active launch site.
- Geographic analysis helped compare launch site distribution.
- Location-based visualization helped identify launch site patterns and supported predictive analysis.

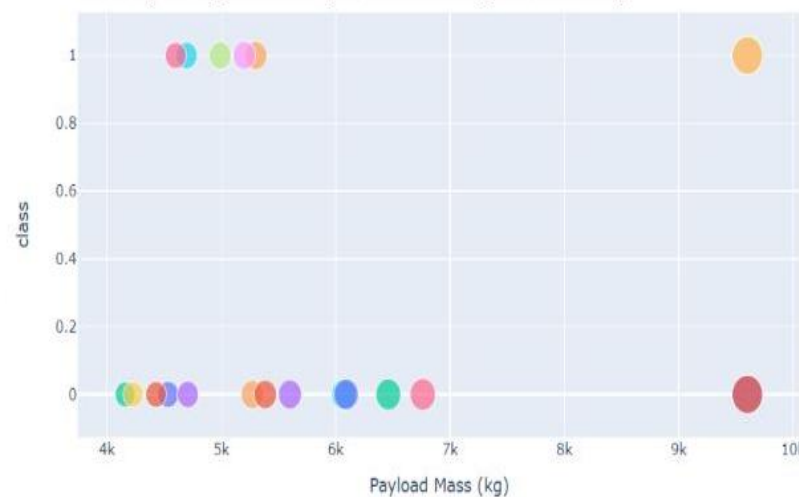
Payload Analysis

Payload vs Launch Success

Low Weighted Payload 0kg – 4000kg



Heavy Weighted Payload 4000kg – 10000kg



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

- **Key Findings:**
- Shows the relationship between payload mass and launch success.
- Interactive payload range slider enables dynamic filtering.
- Different booster versions are represented by different colors.

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Conclusion

• Key Takeaways

- The analysis identified several factors that influence **SpaceX Falcon 9 landing success**.
- **Flight Number** showed a positive relationship with landing success, indicating improved performance over time.
- Payload mass, orbit type, and launch site were key factors influencing landing success.

• Overall Findings

- Landing success rates improved significantly over time, reaching their highest levels by 2019 and remaining consistently high thereafter.
- Historical launch data provides valuable insights into Falcon 9 landing success and supports data-driven decision-making.

GITHUB REPOSITORY



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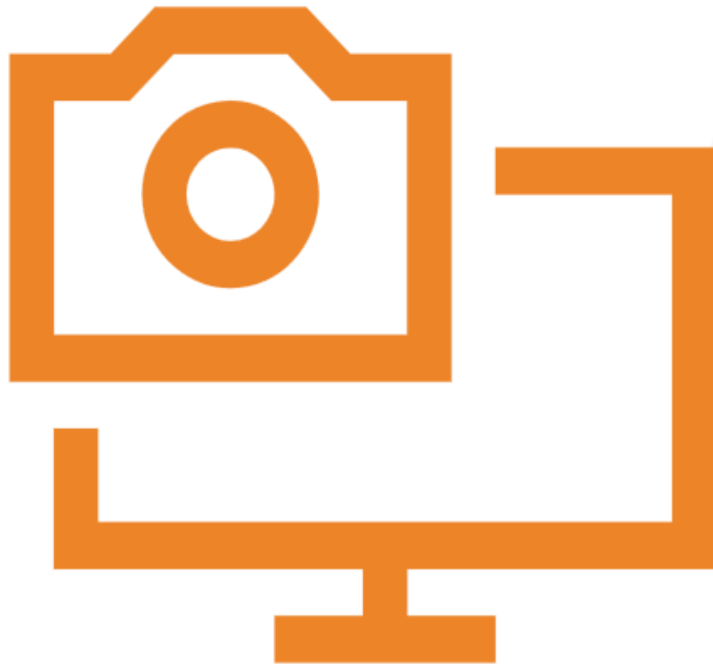
APPENDIX

- **Additional Analysis**

- SQL query outputs used for exploratory analysis.
- Interactive Plotly Dash dashboard for exploratory data analysis.
- Folium map showing launch site proximity.
- Additional EDA charts supporting model development.

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Thank You!

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